

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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Цели и задачи работы

Процесс выполнения лабораторной работы

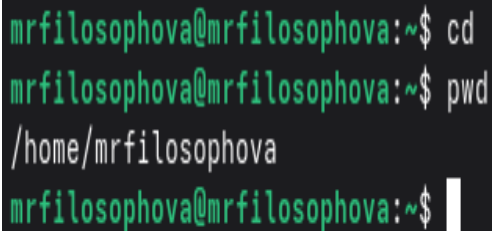
Выводы по проделанной работе

1. Цели и задачи работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

2. Процесс выполнения лабораторной работы



```
mrfilosophova@mrfilosophova:~$ cd  
mrfilosophova@mrfilosophova:~$ pwd  
/home/mrfilosophova  
mrfilosophova@mrfilosophova:~$
```

Рисунок 1: Путь к домашнему каталогу

```
mrfilosophova@mrfilosophova:~$ cd /tmp
mrfilosophova@mrfilosophova:/tmp$ ls
c3a2981b-472f-4945-be2f-17e194344e91.zip
happd.sock
hsperfdata_mrfilosophova
node-compile-cache
par-6d7266696c6f736f70686f7661
snap-private-tmp
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-abrt.service-KT8HbG
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-chronyd.service-hRmaiS
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-colord.service-snQLLl
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-dbus-broker.service-vVx6Rf
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-irqbalance.service-7wDEhb
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-low-memory-monitor.service-rJWLmz
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-ModemManager.service-jQbejG
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-pcsd.service-CxUiXu
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-polkit.service-QLic4Y
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-rtkit-daemon.service-VwfmCz
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-switcheroo-control.service-nlqy5H
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-systemd-logind.service-vTa4Yh
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-systemd-resolved.service-Z9jb6B
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-upower.service-AZRxkL
VMwareDnD
vmware-root_899-3988097379
mrfilosophova@mrfilosophova:/tmp$
```

Рисунок 2: Команда ls

Опции команды ls

```
mrfilosophova@mrfilosophova:/tmp$ ls -la
..
c3a2901b-472f-4945-be2f-17e194344e91.zip
.font-unix
happd.sock
hsperfdata_mrfilosophova
.ICE-unix
node-compile-cache
par-6d7266696c6f736f70686f7661
snap-private-tmp
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-abrt.service-KT8HbG
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-chronyd.service-hRmaIS
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-colord.service-snQlll
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-dbus-broker.service-vVx6Rf
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-irqbalance.service-7wDEhb
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-low-memory-monitor.service-rJWLmz
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-ModemManager.service-jQbejG
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-pcscd.service-CxUiXu
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-polkit.service-QLic4Y
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-rtkit-daemon.service-VwfmCz
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-switcheroo-control.service-nlqy5H
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-systemd-logind.service-vTa4Yh
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-systemd-resolved.service-Z9jb6B
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-upower.service-AZRxkL
VMwareDnD
vmware-root_899-3988097379
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
mrfilosophova@mrfilosophova:/tmp$
```

Рисунок 3: Команда ls -la

Опции команды ls

```
mrfilosophova@mrfilosophova:/tmp$ ls -l
total 3856
-rw-r--r--. 1 mrfilosophova mrfilosophova 3947343 Sep  2 10:31 c3a2901b-472f-4945-be2f-17e194344e91.zip
srw-rw-rw-. 1 root          root          0 Sep  2 10:29 happd.sock
drwxr-xr-x. 2 mrfilosophova mrfilosophova   40 Sep  2 10:43 hspcrfdata_mrfilosophova
drwxr-xr-x. 3 mrfilosophova mrfilosophova   60 Sep  2 10:51 node-compile-cache
drwx-----. 3 mrfilosophova mrfilosophova   60 Sep  2 10:42 par-6d7266696c6f736f70686f7661
drwx-----. 2 root          root          40 Sep  2 10:29 snap-private-tmp
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-abrtd.service-KT8HbG
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-chronyd.service-hRmaiS
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-colord.service-snQlll
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-dbus-broker.service-vVx6Rf
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-irqbalance.service-7wDEhb
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-low-memory-monitor.service-rJWLmz
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-ModemManager.service-jQbejG
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-pcsd.service-CxUiXu
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-polkit.service-QLic4Y
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-rtkit-daemon.service-VwfmCz
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
6ef81-switcheroo-control.service-nlqy5H
drwx-----. 3 root          root          60 Sep  2 10:29 systemd-private-20eb81c712af43ecab45d7a97bb
```

Рисунок 4: Команда ls -l

```
mrfilosophova@mrfilosophova:/tmp$ ls -f
..
node-compile-cache
VMwareDnD
hsperfdata_mrfilosophova
par-6d7266696c6f736f70686f7661
c3a2901b-472f-4945-be2f-17e194344e91.zip
.X1-lock
.X0-lock
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-pcsd.service-CxUiXu
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-colord.service-snQlLl
happd.sock
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-ModemManager.service-jQbejG
vmware-root_899-3988097379
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-systemd-logind.service-vTa4Yh
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-upower.service-AZRxkL
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-switcheroo-control.service-nlqy5H
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-rtkit-daemon.service-VwfmCz
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-polkit.service-QLic4Y
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-low-memory-monitor.service-rJWLmz
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-chrond.service-hRmaiS
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-irqbalance.service-7wDEhb
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-abrt.service-KT8HbG
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-dbus-broker.service-vVx6Rf
systemd-private-20eb81c712af43ecab45d7a97bb6ef81-systemd-resolved.service-Z9jb6B
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
mrfilosophova@mrfilosophova:/tmp$
```

Рисунок 5: Команда ls -f

```
mrfilosophova@mrfilosophova:/tmp$  
mrfilosophova@mrfilosophova:/tmp$ cd /var/spool/  
mrfilosophova@mrfilosophova:/var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 24476 Aug 31 13:37 abrt  
drwx-----. 1 abrt abrt    0 Dec  4 2025 abrt-upload  
drwxr-xr-x. 1 root root   66 Oct 23 2025 anacron  
drwx-----. 1 root root   18 Oct 23 2025 at  
drwx-----. 1 root root    0 Aug  6 2025 cron  
drwx--x---. 1 root lp     6 Dec 12 2025 cups  
drwxr-xr-x. 1 root root    0 Jul 30 2025 lpd  
drwxrwxr-x. 1 root mail  216 Sep  2 10:27 mail  
drwxr-xr-x. 1 root root    0 Jan 27 2026 plymouth  
mrfilosophova@mrfilosophova:/var/spool$
```

Рисунок 6: Каталог /var/spool

```
mrfilosophova@mrfilosophova:/var/spool$ cd
mrfilosophova@mrfilosophova:~$ ls
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work
mrfilosophova@mrfilosophova:~$ ls -al
total 24
drwx-----. 1 mrfilosophova mrfilosophova 370 Sep  2 11:05 .
drwxr-xr-x. 1 root          root          216 Sep  2 10:27 ..
-rw-----. 1 mrfilosophova mrfilosophova 1425 Sep  2 11:07 .bash_history
-rw-r--r--. 1 mrfilosophova mrfilosophova  18 Jul 23  2025 .bash_logout
-rw-r--r--. 1 mrfilosophova mrfilosophova 144 Jul 23  2025 .bash_profile
-rw-r--r--. 1 mrfilosophova mrfilosophova 683 Sep  2 10:51 .bashrc
drwx-----. 1 mrfilosophova mrfilosophova 484 Sep  2 10:58 .cache
drwx-----. 1 mrfilosophova mrfilosophova 360 Sep  2 10:58 .config
drwxr-xr-x. 1 mrfilosophova mrfilosophova   0 Sep  2 10:28 Desktop
drwxr-xr-x. 1 mrfilosophova mrfilosophova   0 Sep  2 10:28 Documents
drwxr-xr-x. 1 mrfilosophova mrfilosophova   0 Sep  2 10:28 Downloads
-rw-r--r--. 1 mrfilosophova mrfilosophova 334 Nov 14  2025 .emacs
-rw-r--r--. 1 mrfilosophova mrfilosophova 233 Sep  2 11:05 .gitconfig
drwxr-xr-x. 1 mrfilosophova mrfilosophova  74 Sep  2 11:02 git-extended
drwx-----. 1 mrfilosophova mrfilosophova 136 Sep  2 10:35 .gnupg
drwx-----. 1 mrfilosophova mrfilosophova  20 Sep  2 10:28 .local
drwxr-xr-x. 1 mrfilosophova mrfilosophova  34 Oct 23  2025 .mozilla
drwxr-xr-x. 1 mrfilosophova mrfilosophova   0 Sep  2 10:28 Music
drwxr-xr-x. 1 mrfilosophova mrfilosophova   0 Sep  2 10:28 Pictures
drwxr-xr-x. 1 mrfilosophova mrfilosophova   0 Sep  2 10:28 Public
drwx-----. 1 mrfilosophova mrfilosophova 132 Sep  2 10:38 .ssh
drwxr-xr-x. 1 mrfilosophova mrfilosophova   0 Sep  2 10:28 Templates
drwxr-xr-x. 1 mrfilosophova mrfilosophova  18 Sep  2 10:42 .texlive2025
drwxr-xr-x. 1 mrfilosophova mrfilosophova   0 Sep  2 10:28 Videos
drwxr-xr-x. 1 mrfilosophova mrfilosophova  10 Sep  2 10:38 work
mrfilosophova@mrfilosophova:~$
```

Рисунок 7: Файлы в домашнем каталоге

```
mrfilosophova@mrfilosophova:~$  
mrfilosophova@mrfilosophova:~$ mkdir newdir  
mrfilosophova@mrfilosophova:~$ mkdir newdir/morefun  
mrfilosophova@mrfilosophova:~$ mkdir letters memos misk  
mrfilosophova@mrfilosophova:~$ ls  
Desktop  Downloads  letters  misk  newdir  Public  Videos  
Documents  git-extended  memos  Music  Pictures  Templates  work  
mrfilosophova@mrfilosophova:~$ rm letters/ memos/ misk/  
rm: cannot remove 'letters/': Is a directory  
rm: cannot remove 'memos/': Is a directory  
rm: cannot remove 'misk/': Is a directory  
mrfilosophova@mrfilosophova:~$ rm -r letters/ memos/ misk/  
mrfilosophova@mrfilosophova:~$ rm -r newdir/  
mrfilosophova@mrfilosophova:~$ ls  
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work  
mrfilosophova@mrfilosophova:~$
```

Рисунок 8: Действия с каталогами

```
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_extensions/ya
madharma/minted-quarto':
_extension.yml  minted-quarto.lua

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':
solvay.jpg

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':
csl  tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/csl
':
gost-r-7-0-5-2008-numeric.csl

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/tex
':
preamble.tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':
image-report  mpv-shot
mrfilosophova@mrfilosophova:~$ ls -t
git-extended  work  Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
mrfilosophova@mrfilosophova:~$
```

Рисунок 9: Команда ls -R и ls -t

```
mrfilosophova@mrfilosophova:~$ help cd
cd: cd [-L|[-P [-e]]] [-@] [dir]
    Change the shell working directory.

Change the current directory to DIR.  The default DIR is the value of the
HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

The variable CDPATH defines the search path for the directory containing
DIR.  Alternative directory names in CDPATH are separated by a colon (:).
A null directory name is the same as the current directory.  If DIR begins
with a slash (/), then CDPATH is not used.

If the directory is not found, and the shell option `cdable_vars' is set,
the word is assumed to be a variable name.  If that variable has a value,
its value is used for DIR.

Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.

Exit Status:
Returns 0 if the directory is changed, and if $PWD is set successfully when
-P is used; non-zero otherwise.
mrfilosophova@mrfilosophova:~$
```

Рисунок 10: Справка по команде cd


```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described here.
    Please refer to your shell's documentation for details about the options it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

COPYRIGHT
    Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes unaffected
        by any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

    --help
        display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

COPYRIGHT
    Copyright © 2025 Free Software Foundation, Inc.  License GPLv3+: GNU GPL version 3 or later
    <https://gnu.org/licenses/gpl.html>.
    Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

```
RM(1)                                User Commands                                RM(1)

NAME
  rm - remove files or directories

SYNOPSIS
  rm [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it
  does not remove directories.

  If the -I or --interactive=once option is given, and there are more than three files or the -r,
-R, or --recursive are given, then rm prompts the user for whether to proceed with the entire op-
  eration.  If the response is not affirmative, the entire command is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is
  not given, or the -i or --interactive=always option is given, rm prompts the user for whether to
  remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
        ignore nonexistent files and arguments, never prompt

  -i
        prompt before every removal

  -I
        prompt once before removing more than three files, or when removing recursively; less in-
        trusive than -i, while still giving protection against most mistakes

  --interactive[=WHEN]
        prompt according to WHEN: never, once (-I), or always (-i); without WHEN, prompt always

  --one-file-system

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

```
122 ls -l
123 cd
124 ls
125 ls -al
126 mkdir newdir
127 mkdir newdir/morefun
128 mkdir letters memos misk
129 ls
130 rm letters/ memos/ misk/
131 rm -r letters/ memos/ misk/
132 rm -r newdir/
133 ls
134 ls -R
135 ls -t
136 help cd
137 man pwd
138 man mkdir
139 man rmdir
140 man rm
141 history
mrfilosophova@mrfilosophova:~$
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.